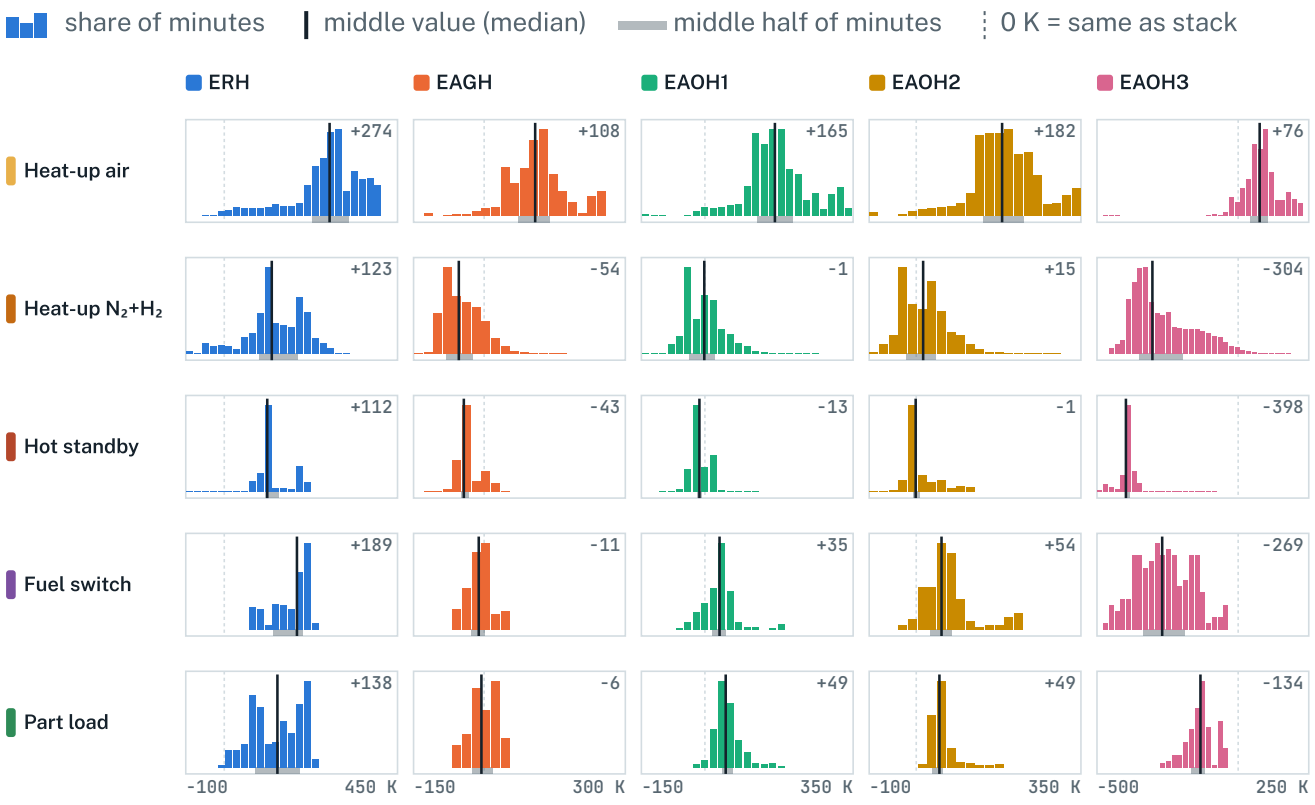
HEATERS VS STAGES · FOLLOW-UP ON “HEATER MINUS STACK”

Each number sits on top of a spread

Yes, every number in the “heater minus stack” table comes from a distribution. Behind each one are hundreds to about 20,000 one-minute readings. The table shows only the middle of that spread. Some spreads are tight, so the middle is a fair summary. Others are wide or have two humps, so one number hides a lot.



One small chart per stage (rows) and heater (columns): how often each difference between heater and stack occurred, minute by minute, while the heaters were on. Bars right of the dashed line mean the heater pipe was hotter than the stack. Hover a chart for its numbers.

What the shapes say

- **Hot standby is tight.** For EAGH, EAOH1, EAOH2 and EAOH3 half of all minutes fall inside a band of 12–17 K. Here the table number is a reliable summary.
- **The heat-up stages are wide** (middle half spans 57–153 K). The stack is still warming and the pipes warm at a different pace, so the gap keeps changing through the stage.

- **Some are lopsided**, with a long tail to one side: ERH in Heat-up air, EAOH3 in Heat-up air, EAOH1 in Heat-up N_2+H_2, EAOH2 in Heat-up N_2+H_2, EAOH1 in Fuel switch, EAOH1 in Part load.
- **Some have two humps**, which means two different situations are mixed in one stage: EAOH3 in Heat-up N_2+H_2, EAOH2 in Hot standby, ERH in Fuel switch, EAOH2 in Fuel switch, ERH in Part load, EAOH2 in Part load. The next page shows where the two humps come from.

A better table: middle value with its typical range

Each cell shows the middle minute (median) and, below it, the range that holds the middle half of the minutes. Values in kelvin; one kelvin is the same step as one degree Celsius.

STAGE	ERH	EAGH	EAOH1	EAOH2	EAOH3
Heat-up air 2,415 min · 12 runs	+274 +228 … +322	+108 +72 … +140	+165 +124 … +208	+182 +143 … +228	+76 +43 … +104
Heat-up N_2+H_2 4,443 min · 12 runs	+123 +91 … +191	-54 -80 … -23	-1 -38 … +23	+15 -23 … +41	-304 -351 … -198
Hot standby 19,725 min · 21 runs	+112 +106 … +140	-43 -49 … -32	-13 -20 … -8	-1 -6 … +8	-398 -400 … -383
Fuel switch 168 min · 12 runs	+189 +126 … +203	-11 -27 … +2	+35 +18 … +49	+54 +30 … +76	-269 -338 … -191
Part load 167 min · 8 runs	+138 +79 … +195	-6 -25 … +18	+49 +40 … +64	+49 +35 … +57	-134 -165 … -118

Small correction. The earlier table took the middle heater temperature and subtracted the middle stack temperature. This table takes the middle of the minute-by-minute differences, which is the more correct way. They mostly agree, but can differ by up to 43 K (ERH in Part load: +181 before, +138 now).

> **If you remember one thing:** trust the hot-standby numbers; read the heat-up, fuel-switch and part-load numbers as a rough middle of a wide spread.

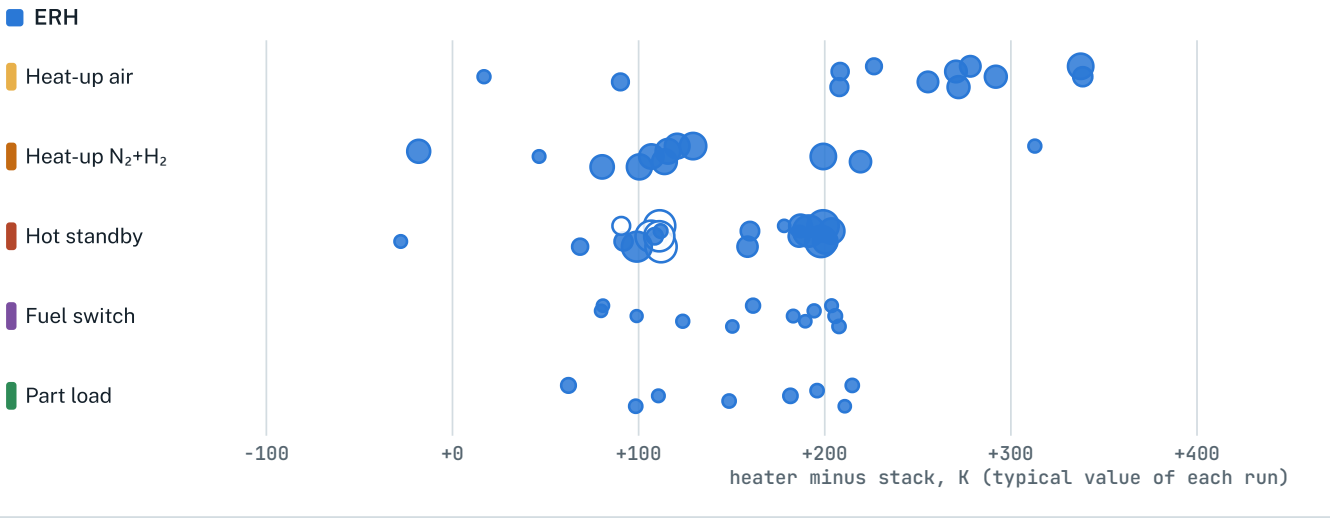
HEATERS VS STAGES · FOLLOW-UP, PAGE 2

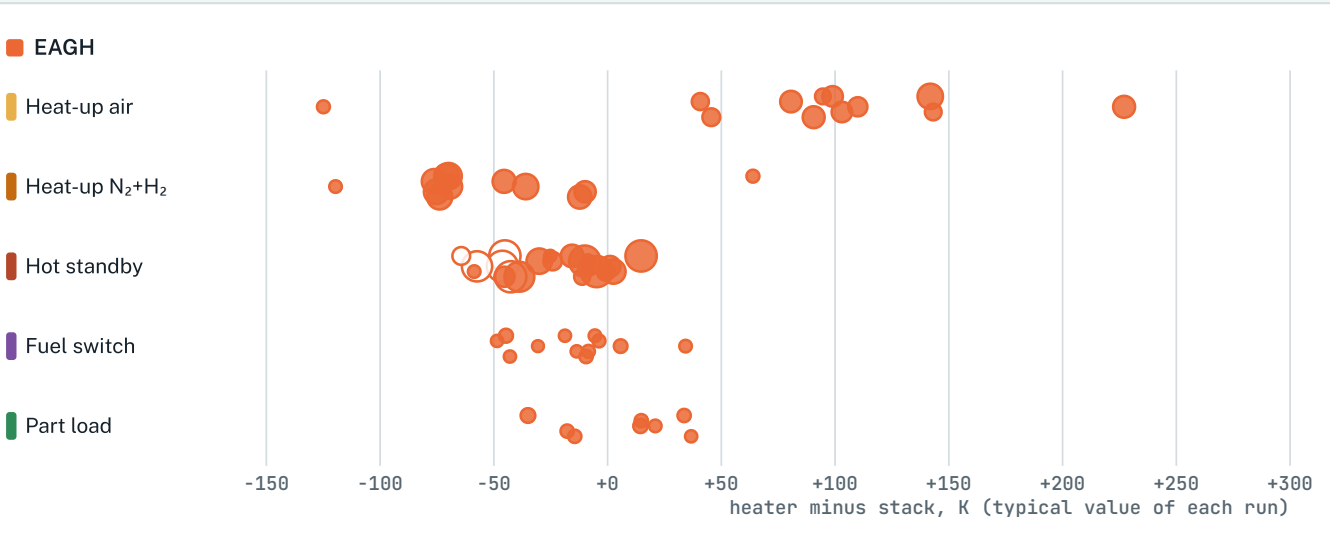
The runs do not all look alike

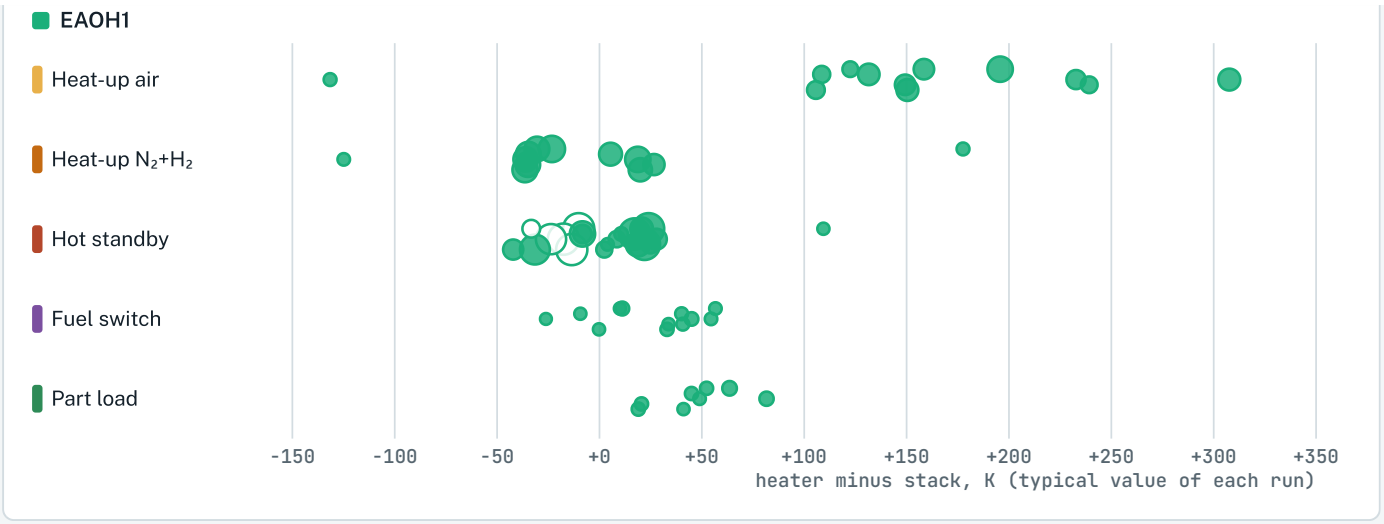
Minutes inside one run are not independent: a heater that runs 100 K hot at 10:00 is still about 100 K hot at 10:01. So the honest sample size is the **number of runs** (8–21 per stage), not the thousands of minutes.

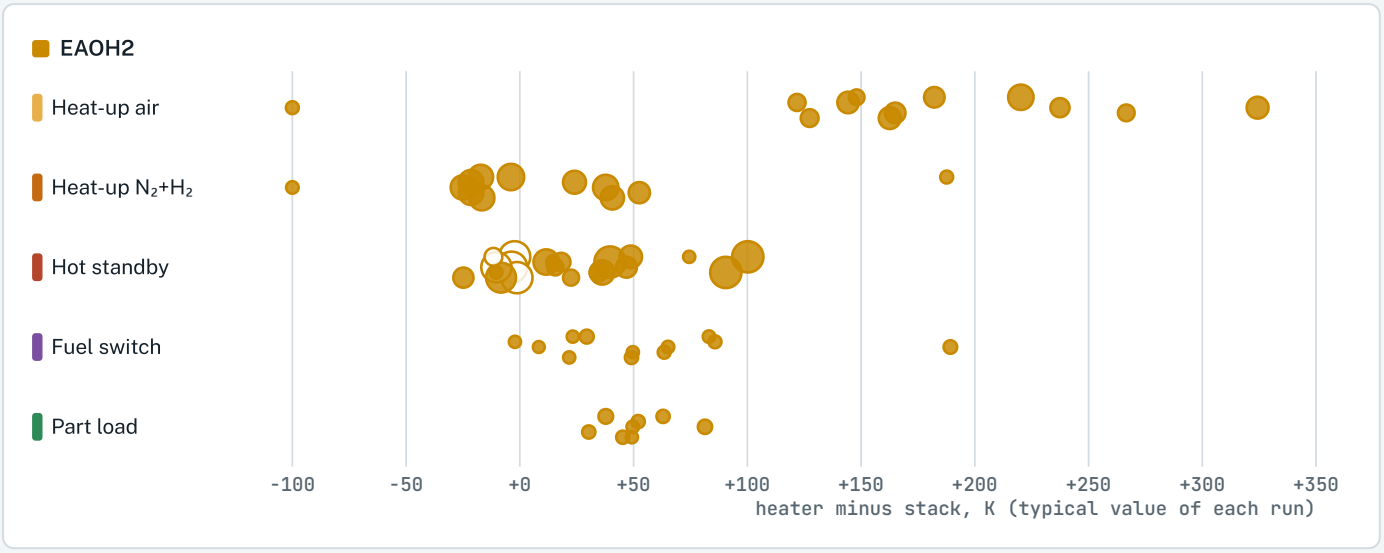
Each dot below is one run of a stage. It sits at that run's typical (median) heater-minus-stack value. Bigger dots are longer runs. Hover a dot for its date and length.

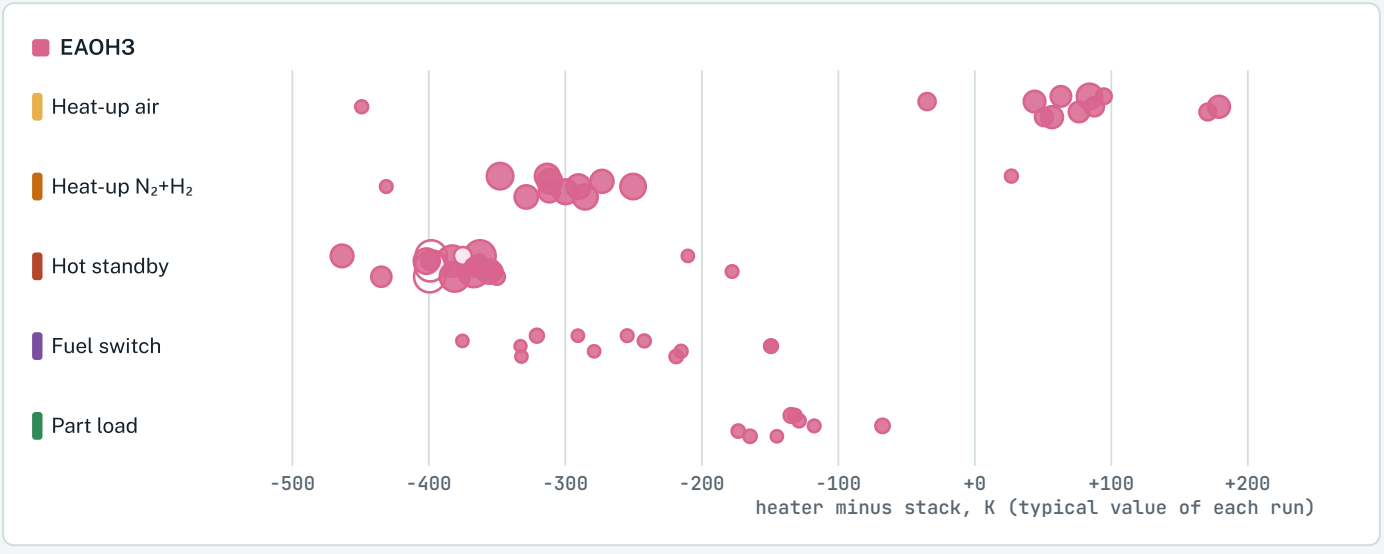
● run ○ hot standby held at about 510 °C (stack below 550 °C)











Where the two humps come from

- **Hot standby has two kinds of run.** The 5 runs held at about 510 °C (31 Jan, 14 Feb, 18 Feb, 01 Mar, 29 Mar) give ERH about +111 K. The 16 hotter standbys near 601 °C, all in March, give ERH about +169 K. ERH is driven to about 700 °C there instead of about

620 °C. This split widens ERH's spread and gives EAOH2 its two humps on page 1. Four of the hold runs last from 15 hours to about 3 days, so they dominate the minute count.

- **Heat-up air has one odd run**: 31 Jan 17:25, 13 min, with the heaters just switching (ERH +17 K, EAOH3 −449 K). Very short runs like this sit at the edges of every row.
- **Fuel switch and part load** have only 8–12 runs of 5–43 minutes. ERH varies from about +60 to +215 K between runs, so their table numbers rest on few, short runs.
- **EAOH3 in hot standby** is the steadiest value in the data: almost every run sits between −400 and −350 K.

What to do with this

1. Show each table number together with its range (page 1), not on its own.
2. Split hot standby into the ~510 °C hold and the ~600 °C standby when describing heater behaviour, or at least report both values.
3. Treat fuel switch and part load numbers as provisional until more runs are logged.

> **If you remember one thing:** count runs, not minutes. Hot standby is really two groups, and the short stages rest on 8–12 runs.